AMERICAN ACADEMY OF OPHTHALMOLOGY®

The Noninvasive Measurement of Ocular Rigidity and the Pulsatile Choroidal Volume Change in Children and Young Adults

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Purpose: Measuring ocular rigidity (OR) and the pulsatile choroidal volume change (ΔV) from childhood to adulthood would provide essential insight into the role of the sclera and choroid in axial myopia. A validated measurement method based on Beaton et al (2015) is the only noninvasive, direct, and reliable method available, yet it has never been used in children. This study aims to assess the feasibility and repeatability of noninvasive OR and ΔV measurements in children and young adults using this method.

Design: This is a cross-sectional study.

Subjects: Children and young adults aged 6 to 26 years.

Methods: OCT videos were acquired using enhanced depth imaging mode. A neural network (NN) approach was used to extract the choroid segmentation from OCT videos, which was then used to measure choroidal filling as described by Beaton et al. Ocular rigidity was computed from the ΔV and the ocular pulse amplitude using Friedenwald equation. Intrasession repeatability was assessed for subjects with 2 consecutive measurements of OR.

Main Outcome Measures: Ocular rigidity, pulsatile choroidal volume change, and the method's yield.

Results: Sixty-seven subjects (67 eyes, 27 males) aged 13 ± 6 years were enrolled, and 62 subjects completed the study. ΔV and OR were computed. Out of the 62 videos, 98% (61) of the OCT videos were successfully segmented using the NN approach, with heart rate detectable in 79% (48) of the videos. Average OR and ΔV were 0.027 ± 0.022 μL^{-1} and 5.7 ± 2.8 μL, respectively (n = 48). The mean submacular choroidal thickness (CT) and pulsatile CT change were 283.6 ± 40.4 μm and 8.36 ± 4.4 μm, respectively. Intrasession repeatability for OR and ΔV was assessed in 31 eyes and was determined to be excellent based on a single measure intraclass correlation coefficient of 0.912, 95% confidence interval (CI) (0.825, 0.956) and 0.941, 95% CI (0.879, 0.972), respectively and a within-subject standard deviation of 0.0037 μL^{-1} and 1.14 μL, respectively.

Conclusions: Measures of OR and ΔV are achievable in children and young adults and have good repeatability. This noninvasive method can be used to establish structural biomechanical changes of the sclera and choroid with myopic axial elongation in childhood and beyond.

Financial Disclosure(s): The author(s) have no proprietary or commercial interest in any materials discussed in this article. *Ophthalmology Science 2025;5:100814* © *2025 by the American Academy of Ophthalmology. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).*

Myopia develops in childhood and early adolescence and its prevalence has been increasing to epidemic proportions. It is expected that 50% of the global population will be myopic by 2050.[1] In some urbanized countries, 90% of young adults have been diagnosed with myopia.[2] This is a public health problem since as many as 20% of high myopes suffer severe complications leading to visual impairment and blindness and significant socioeconomic burden.[3–5] Although the mechanisms regulating eye growth are thought to be multifactorial, much remains unknown about the pathophysiology of myopia. There is evidence from experimental data that biomechanical properties of the sclera may play an important role in myopic axial elongation.[6–9] However, this needs to be clinically investigated.

Ocular rigidity (OR) is defined as the resistance that the eye globe exerts against distending forces and is an important biomechanical parameter of the eye. The primary contributor to OR is scleral stiffness.[10,11] Ocular rigidity is estimated using the Friedenwald function[10] which relates intraocular pressure (IOP) and volume (V) changes in the eye as follows:

$$\ln \frac{IOP}{IOP_0} = OR * (V - V_0)$$

A greater OR value represents a rigid eye or stiffer sclera (akin to a soccer ball), whereas a low OR value represents a more compliant eye (akin to a birthday balloon).

To measure OR, anterior chamber manometry is the gold standard method, in which a known V of fluid is injected in the eye and the corresponding IOP change is measured. However,

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https://doi.org/10.1016/j.xops.2025.100814
ISSN 2666-9145/25

this method is invasive and thus clinically unpractical.[12] Previous attempts to measure OR noninvasively using differential tonometry were found to be inaccurate.[13–15] A direct and noninvasive method has been developed to measure OR using dynamic OCT imaging of the choroid and tonometry.[16] Instead of injecting fluid in the eye using a needle,[12,17] the innovative method developed by Beaton et al[16] involves measuring the pulsatile choroidal volume change (ΔV), or the V of blood entering the eye with each heartbeat, using video-rate OCT imaging of the choroid coupled with an automated segmentation algorithm. The corresponding pulsatile IOP change is then obtained using dynamic contour tonometry, and OR is computed. This method, which has been validated against an invasive procedure,[18] has since been used to investigate OR in adults with open-angle glaucoma [11,19–22] and space-flight associated neuro-ocular syndrome.[23]

Because of the increasing prevalence of myopia and risk of visual impairment with high myopia, great effort is made to prevent its progression during childhood.[24] Ocular rigidity may be a modifiable parameter and the target of novel therapies that would stiffen the sclera to prevent further ocular growth.[25] Evidence supports an association between low OR and axially elongated eyes.[10,26,27] In fact, lower OR was found in older adult eyes with established axial myopia compared with nonmyopic eyes (0.015/μL vs. 0.033/μL, $P < 0.001$) using this OCT-based method.[28] Despite the noninvasive nature of the method, it has not been used in children or young adults and the feasibility of OR measurements in children has yet to be assessed. Ocular rigidity may be a biomarker for predicting and mitigating myopia onset and progression, but also for monitoring scleral alterations in myopia and the effectiveness of multifocal lenses, atropine, and other myopia control modalities.[29] Furthermore, recent findings suggest an association between OR and choroidal blood flow,[22] warranting the need to further investigate the link between the sclera and choroid in axial myopia.

Therefore, the purpose of this study is to assess for the first time the feasibility and repeatability of noninvasive OR and ΔV measurements in children and young adults using dynamic OCT imaging and tonometry.

Methods

This study followed the tenets of the Declaration of Helsinki and was reviewed by the University of Houston institutional review board. Institutional review board approval was obtained. Before testing, informed consent was obtained from all participants, and parental permission and child assent were obtained for minor subjects.

Children and young adults aged 6 to 26 years were recruited from the University Eye Institute at the University of Houston College of Optometry. Participants were required to have clear ocular media. Patients with a previous history of eye injury, eye trauma, narrow angles, nystagmus, or intraocular surgery were excluded. The study visit included video-rate OCT imaging (Spectralis Spectral Domain OCT, Heidelberg Engineering GmbH) and simultaneous finger oximetry, IOP measurements using dynamic contour tonometry (Pascal, Ziemer Ophthalmic Systems AG), and ocular axial length (AL) and anterior chamber depth (ACD) measurement (Lenstar LS900, Haagstreit). The study eye was chosen randomly. For a subset of subjects, OCT videos, oximetry, dynamic contour tonometry, and biometry measurements were successively repeated twice on the same eye within the same session to assess repeatability.

A single line 30° wide B-scan was centered on the fovea and the azimuthal angle was chosen to optimize the visibility of the choroid-sclera interface. OCT videos were acquired with high-speed at a frame rate of ≥9 Hz, which is above the Nyquist sampling rate, and with enhanced depth imaging modes. Each video contained 1024 B-scans with image resolution of 768 × 496 pixels. The heart rate was measured simultaneously using finger oximetry. Analysis of the videos was then carried out using a neural network (NN) trained on 10 798 manually segmented macular OCT B-scans acquired using the same OCT device. Most of the B-scans were centered on the macular region, and the B-scans were of different dimensions. The network was trained for 10 epochs, to an accuracy of 99.25% and cross-entropy loss of 0.023, using a Deeplab v3+ framework and Resnet 50 in Matlab (The MathWorks Inc).[30] The mean Intersection over Union was 0.9823, and the weighted Intersection over Union was 0.9847. While originally designed for segmentation of static images, the NN was adapted to carry out choroidal time-series segmentation based on the methods described in detail in Beaton et al.[16] Briefly, the choroid's boundaries, i.e., Bruch's membrane and the choroid-sclera interface, are delineated in each frame of the video using the NN (see Fig 1). If any A-scans contain no choroid (e.g., around the optic disc), these A-scans can be

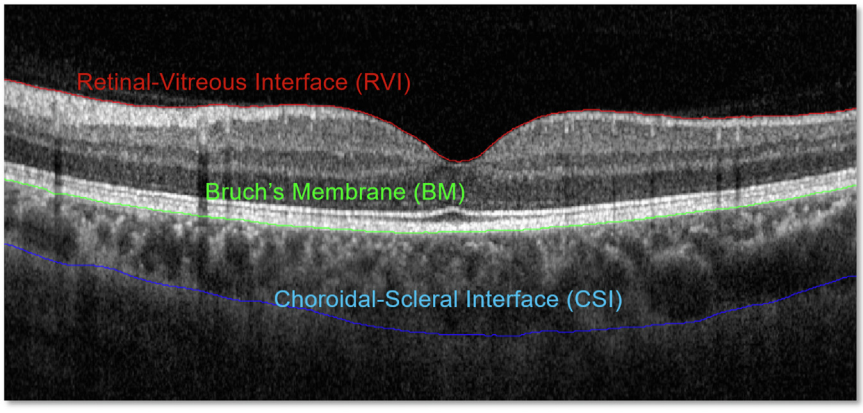


Figure 1. Segmentation of a single B-scan from an OCT video. Choroidal thickness corresponds to the distance between BM and the CSI. The pulsatile CT change (ΔCT) is computed as the CT change between each frame of the video. BM = Bruch's membrane; CSI = choroidal-scleral interface; CT = choroidal thickness.

trimmed from the edges of the B-scan. Similarly, B-scans with a Q score <20 (Spectralis quality parameter) are discarded. The mean choroidal thickness (CT) is then computed for each B-scan as the mean of the vertical distance between the Bruch's membrane and choroid-sclera interface. After further postprocessing, which includes removal of B-scans with CT <39 μm or CT >585 μm, the pulsatile CT change (ΔCT) is obtained throughout the time series. Since the choroid accounts for approximately 90% of the total ocular blood flow, we extrapolate the pulsatile choroidal volume change (ΔV) from ΔCT using a two-sphere mathematical model of the choroid adapted from our previous work[18]:

$$\Delta V \cong \frac{\pi}{2}\left(AL_{adj} + CT\right)^2 \Delta CT$$

where $AL_{adjusted}$ is equal to AL - ACD. A frequency spectrum analysis was performed to ensure that CT fluctuations in the OCT videos are due to the pulsatile blood flow (see Fig 2). High frequency components needed to coincide with the heart rate frequency measured simultaneously using a finger oximeter for the results to be considered "valid."[16] As described in Beaton et al,[16] to extract the CT fluctuations associated mainly to the heart rate and discard respiration, head movement, saccades, and segmentation noise, frequency components below half of the heart rate frequency, >3 times the heart rate frequency, and those with values <10% of the maximum peak within this range were filtered out. Immediately after imaging, 1 drop of anesthetic (proparacaine HCl 0.5%) was instilled in the study eye, and the IOP and ocular pulse amplitude (OPA) were measured using the Pascal tonometer, which is independent of corneal biomechanical properties.[31–33] The IOP is the diastolic IOP and the OPA is the amplitude of the difference between the systolic and diastolic IOP. The pulsatile IOP change or IOP/IOP_0 in the Friedenwald function[10] is defined as $(IOP+OPA)/IOP$. Finally, OR was computed from the V and IOP changes ensued in the eye between the systole and diastole.

The yield was defined as the number of "valid' videos (in which the heart rate was found in the frequency of choroidal pulsatility) over the total number of videos analyzed.

To assess repeatability, 2 consecutive measurements were carried out within the same session. Data analysis and assessment of repeatability was conducted using paired *t* test, Bland−Altman plot, the within-subject standard deviation,[34] and the intraclass correlation coefficient (ICC) calculated within an absolute agreement, 2-way mixed-effects model using Stata statistical software (StataCorp LLC). For all statistical tests, *P* values <0.05 were considered significant.

Results

Sixty-seven subjects (67 eyes, 27 males) aged 13 ± 6 years were enrolled, and 62 subjects (93%) completed the study. Out of the 62 videos, 98% (61) of the OCT videos were successfully segmented using the NN approach we adapted for time series; 79% (48) of videos were considered "valid" based on the frequency analysis corresponding with the heart rate.

The demographic and clinical characteristics of this cohort of children and young adults, including 25 myopes, is described in Table 1. The mean submacular CT was

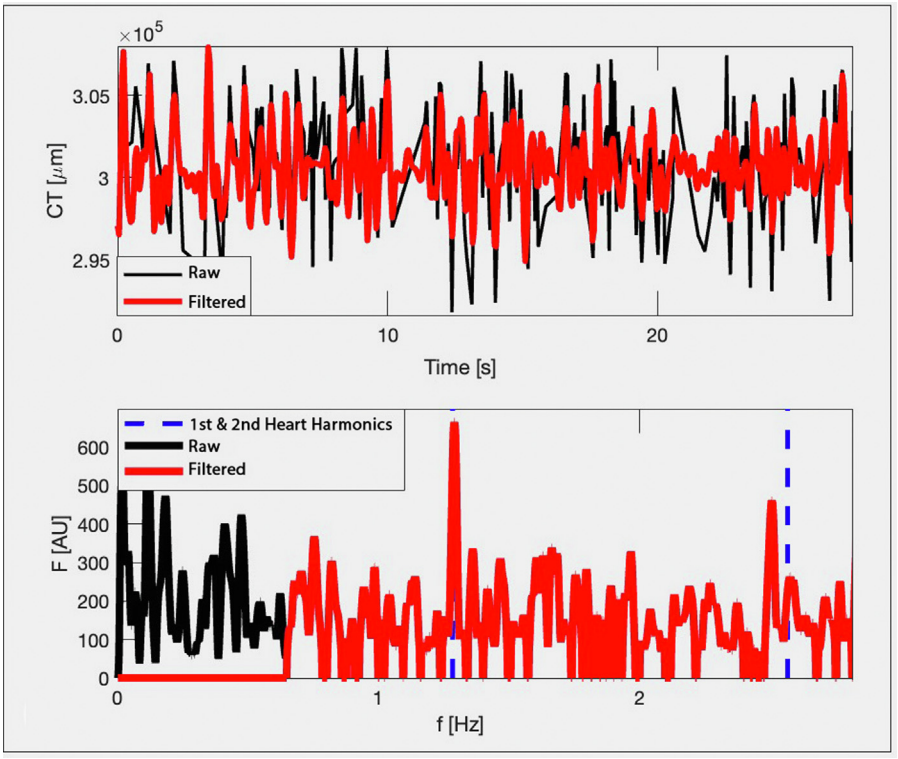


Figure 2. (Top) Frequency spectrum analysis of CT fluctuations with respect to time showing the raw fluctuations of CT versus time (black) and band-pass filtered CT signal (red). (Bottom) The dashed blue lines indicate the first 2 harmonics of the measured heart rate. The frequency spectrum analysis shows correspondence with the heart rate frequency, confirming that fluctuations in CT with time are due to the pulsatile blood flow. AU = arbitrary units; CT = choroidal thickness.

283.6 ± 40.4 μm, the ΔCT throughout the videos was 8.36 ± 4.4 μm, the ΔV was 5.7 ± 2.8 μL, and the OR was 0.027 ± 0.022 μL^{-1}. The Pascal-IOP and OPA were 14.8 ± 2.3 mmHg and 1.9 ± 1.1 mmHg, respectively, the AL was 23.93 ± 1.23 mm, and the ACD was 3.62 ± 0.29 mm. By multiplying ΔV by the heart rate for each subject, the average pulsatile ocular blood flow was estimated to be 498.3 ± 280.8 μL/minute.

In terms of feasibility, only 5 subjects (4 males, 9 ± 2 years old) out of 67 did not complete the study; 2 subjects had difficulty with adequate target fixation and being still during testing, and 3 subjects withdrew due to fear or anxiety from the proximity of contact tonometry.

To assess intrasession repeatability, 31 subjects (31 eyes, 12 males) aged 14 ± 7 years had all measurements repeated 2 times, including Pascal tonometry, ocular biometry, and OCT videos. No difference was found between the repeated measures for IOP, OPA, AL, ACD, ΔV, and OR with paired *t* test ($P \geq 0.05$). Good test-retest reliability was found for OR and ΔV obtained using the NN approach in 31 eyes, as shown in Table 2. The within-subject standard deviation was 0.0037 μL^{-1} and 1.14 μL for OR and ΔV, respectively. The single measure ICC was 0.912, 95% confidence interval (CI) (0.825, 0.956) for OR and 0.941, 95% CI (0.879, 0.972) for ΔV. The average measures ICC for OR and ΔV was 0.954, 95% CI (0.904, 0.978) and 0.970, 95% CI (0.935, 0.986), respectively. The concordance correlation coefficient was 0.939 and 0.951 for OR and ΔV, respectively, indicating strong agreement between repeated measurements (Table 2). The Bland–Altman plots also showed good agreement between both measurements of OR (0.00022/μL, 95% limit of agreement [−0.01009, 0.01052]) and ΔV (0.053 μL, 95% limit of agreement [−2.523, 3.576]) (Fig 3). The mean absolute difference between repeated measurements was 0.004/μL and 1.23 μL for OR and ΔV, respectively.

Table 1. Demographic and Clinical Characteristics for 48 Children and Young Adults (48 Eyes) Who Completed the Study and Obtained Valid Ocular Rigidity Measurements

Variables	Values
Eye (OD)	22 (46%)
Sex (M)	14 (29%)
Age (years)	14 ± 6
Spherical equivalent refraction (D)	−1.41 ± 2.45
Number of myopic eyes (SER ≤−0.50 D)	25 (52%)
Anterior chamber depth (mm)	3.62 ± 0.29
Axial length (mm)	23.93 ± 1.23
Pascal-IOP (mmHg)	14.8 ± 2.3
Ocular pulse amplitude (mmHg)	1.9 ± 1.1
Mean submacular choroidal thickness (μm)	283.6 ± 40.4
Pulsatile choroidal thickness change (μm)	8.36 ± 4.4
Pulsatile choroidal volume change (μL)	5.7 ± 2.8
Ocular rigidity (μL^{-1})	0.027 ± 0.022

D = diopters; IOP = intraocular pressure; M = male; OD = right eye; SER = spherical equivalent refraction.
The data are presented as mean ± standard deviation where applicable.

Table 2. Repeatability of OR, the ΔV, and Other Repeated Measures Obtained Using the Neural Network Approach in 31 Subjects (31 Eyes) Aged 6 to 26 Yrs

Variables	Average ± SD	S_w	Single Measures ICC	Average Measures ICC	CCC	Mean Difference	Mean Absolute Difference	95% LOA
IOP (mmHg)	15.6 ± 2.3	1.03	0.811	0.895	0.806	0.158	1.14	−2.72, 3.04
OPA (mmHg)	1.6 ± 0.6	0.25	0.822	0.902	0.817	0.045	0.28	−0.65, 0.74
AL (mm)	23.90 ± 1.11	0.012	0.999	0.999	1.000	0.001	0.012	−0.034, 0.032
ACD (mm)	3.68 ± 0.25	0.018	0.995	0.997	0.994	0.006	0.016	−0.044, 0.056
ΔV (μL)	7.05 ± 4.65	1.14	0.941	0.970	0.939	0.53	1.23	−2.52, 3.58
OR (μL^{-1})	0.019 ± 0.012	0.0037	0.912	0.954	0.909	0.00022	0.0036	−0.010, 0.011

ACD = anterior chamber depth; AL = axial length; CCC = concordance correlation coefficient; ICC = intraclass correlation coefficient; IOP = intraocular pressure; LOA = limits of agreement; OPA = ocular pulse amplitude; OR = ocular rigidity; SD = standard deviation; S_w = within-subject standard deviation; ΔV = pulsatile choroidal volume change.

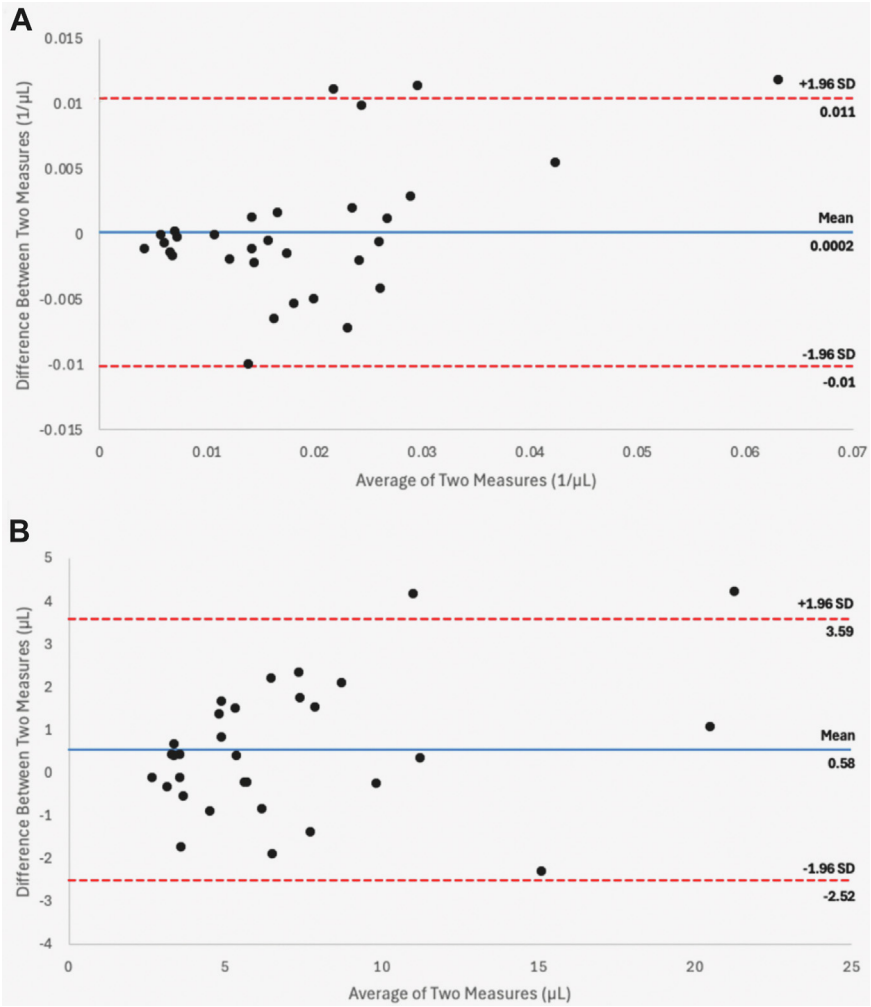


Figure 3. Bland–Altman plots showing agreement between 2 repeated measurements of (**A**) ocular rigidity (μL^{-1}) and (**B**) pulsatile choroidal volume change (μL) in 31 children and young adults. The x-axis represents the average of the 2 measurements while the y-axis represents the difference between the first and second measurements. The central horizontal blue line corresponds to the mean of the difference. The upper and lower horizontal red lines correspond to the upper and lower bounds of the 95% confidence interval of the difference between the first and second measurement. SD = standard deviation.

Discussion

In this study, we show the feasibility of measuring ΔV and OR in children and young adults aged 6 to 26 years. Most children were able to sit for the testing, and the NN-based approach for OCT segmentation was effective for measuring the pulsatile choroidal volume change. With the combination of ΔV obtained using OCT videos and the pulsatile IOP change obtained using Pascal tonometry, OR could be estimated using Friedenwald's equation. The high yield suggests that these methods can be used to non-invasively assess ocular biomechanics in young eyes.

In terms of feasibility, while some challenges were expected in the application of this method to children and young adults (since a steady fixation during the videos is required and the method involves contact tonometry), a high number (93%) of young subjects were able to complete the study. A few strategies we utilized during study visits included repeating positive instructions during testing, using encouragement and updating the child on the progress made during each test, and storytelling during some of the measurements. The 5 children who did not complete the study had difficulty with steady fixation, often related to unmedicated attention deficit hyperactivity disorder at the time of the study visit or anxiety, which could be further improved in the future by using techniques such as music or play interventions.[35,36]

Choroid segmentation was successfully carried out in all except 1 (98%) of the videos using the NN approach adapted for time series, despite low image averaging rate since OCT videos need to be acquired at a faster rate than the heart rate to capture pulsatile choroidal fluctuations. The heart rate was detected in 48 of 61 videos, resulting in a 79% yield for this method, which is greater than previous reports.[20] The results of this study are thus extremely promising in confirming the ability to lead clinical studies

on OR, the ΔV, and their role in childhood myopia and other relevant eye diseases.

We provide the first description of the ΔV (5.7 ± 2.8 μL) and the OR (0.027 ± 0.022 μL^{-1}) in a population of children and young adults. Other parameters of interest include the ΔCT which was 8.36 ± 4.4 μm in this cohort. This parameter indicates the magnitude of CT change that occurs with each heartbeat. We believe that ΔCT is an essential parameter that should be carefully considered in most myopia studies investigating CT variation to correctly interpret results in terms of test–retest (or longitudinal data) variability.

Intrasession repeatability was assessed in 31 eyes from 31 subjects aged 14 ± 7 years. All repeated measures for IOP, OPA, AL, ACD, ΔV, and OR were found not to be significantly different ($P \geq 0.05$), and good test–retest reliability was found for these parameters as indicated by the ICC and within-subject standard deviation. The concordance correlation coefficient indicated strong agreement between repeated measurements. The Bland–Altman plots also showed good agreement between both measurements of OR and ΔV, indicating good repeatability. The Bland–Altman plots for both OR and ΔV showed a small mean difference as well as narrow 95% limit of agreement between repeated measurements. Together with the high yield, the excellent repeatability suggests that these metrics could be used to monitor changes in OR and ΔV in longitudinal studies of myopia progression.

There are limitations to this study. Since there is an assumption of sphericity in the choroidal model used to derive the pulsatile choroidal volume change from the pulsatile CT change, it should therefore not be applied in eyes that present with a staphyloma or other shape irregularities. In addition, OCT images were not corrected for posterior eye shape[37] before measuring CT, which may influence estimated volumes. Moreover, we did not account for positional differences during acquisition between the 2 repeated measurements based on Listing's law.

Future work will focus on measuring OR and ΔV to investigate the role of the sclera and choroid in childhood ocular diseases, particularly axial myopia. Furthermore, since the method uses equipment that is commonly used in most clinical settings, it could be developed into a screening tool in clinical settings.

Footnotes and Disclosures

Originally received: September 20, 2024.
Final revision: March 30, 2025.
Accepted: April 22, 2025.
Available online: April 26, 2025. Manuscript no. XOPS-D-24-00388.

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Disclosure(s):

The Article Publishing Charge (APC) for this article was paid by American Academy of Optometry Foundation.

All authors have completed and submitted the ICMJE disclosures form.

The authors have no proprietary or commercial interest in any materials discussed in this article.

HUMAN SUBJECTS: Human subjects were included in this study. This study followed the tenets of the Declaration of Helsinki and was reviewed by the University of Houston institutional review board. Institutional review board approval was obtained. Before testing, informed consent was obtained from all participants, and parental permission and child assent were obtained for minor subjects.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Sayah

Data collection: Reyes, Smith, Lam, Adel, Sayah

Analysis and interpretation: Reyes, Smith, Patel, Sayah

Obtained funding: Sayah

Overall responsibility: Reyes, Smith, Lam, Adel, Patel, Sayah

Abbreviations and Acronyms:
ACD = anterior chamber depth; **AL** = axial length; **CI** = confidence interval; **CT** = choroidal thickness; **ICC** = intraclass correlation coefficient; **IOP** = intraocular pressure; **NN** = neural network; **OPA** = ocular pulse amplitude; **OR** = ocular rigidity; **V** = volume.

Keywords:
Biomechanics, Children, Choroid, Myopia, Ocular rigidity.

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References

1. Holden BA, Fricke TR, Wilson DA, et al. Global prevalence of myopia and high myopia and temporal trends from 2000 through 2050. *Ophthalmology*. 2016;123:1036–1042.
2. Morgan IG, French AN, Ashby RS, et al. The epidemics of myopia: aetiology and prevention. *Prog Retin Eye Res*. 2018;62:134–149.
3. Ohno-Matsui K, Lai TY, Lai CC, Cheung CM. Updates of pathologic myopia. *Prog Retin Eye Res*. 2016;52:156–187.
4. *The Impact of Myopia and High Myopia*. Sydney, Australia: World Heath Organization - Brien Holden Vision Institute; 2015.
5. Saw SM, Gazzard G, Shih-Yen EC, Chua WH. Myopia and associated pathological complications. *Ophthalmic Physiol Opt*. 2005;25:381–391.
6. Grytz R. Scleral remodeling in myopia. In: Roberts C, Dupps WJ, Downs JC, eds. *Biomechanics of the Eye*.

Amsterdam, The Netherlands: Kugler Publications; 2018: 383–403.
7. McBrien NA, Norton TT. Prevention of collagen crosslinking increases form-deprivation myopia in tree shrew. *Exp Eye Res*. 1994;59:475–486.
8. McBrien NA, Gentle A. Role of the sclera in the development and pathological complications of myopia. *Prog Retin Eye Res*. 2003;22:307–338.
9. Phillips JR, Khalaj M, McBrien NA. Induced myopia associated with increased scleral creep in chick and tree shrew eyes. *Invest Ophthalmol Vis Sci*. 2000;41:2028–2034.
10. Friedenwald JS. Contribution to the theory and practice of tonometry. *Am J Ophthalmol*. 1937;20:985–1024.
11. Ma Y, Moroi SE, Roberts CJ. Non-invasive clinical measurement of ocular rigidity and comparison to biomechanical and morphological parameters in glaucomatous and healthy subjects. *Front Med (Lausanne)*. 2021;8:701997.
12. Pallikaris IG, Kymionis GD, Ginis HS, et al. Ocular rigidity in living human eyes. *Invest Ophthalmol Vis Sci*. 2005;46: 409–414.
13. Pallikaris IG, Dastiridou AI, Tsilimbaris MK, et al. Ocular rigidity. *Expert Rev Ophthalmol*. 2010;5:343–351.
14. Moses RA, Grodzki WJ. Ocular rigidity in tonography. *Doc Ophthalmol*. 1969;26:118–129.
15. Jackson CR. Schiotz tonometers. An assessment of their usefulness. *Br J Ophthalmol*. 1965;49:478–484.
16. Beaton L, Mazzaferri J, Lalonde F, et al. Non-invasive measurement of choroidal volume change and ocular rigidity through automated segmentation of high-speed OCT imaging. *Biomed Opt Express*. 2015;6:1694–1706.
17. Eisenlohr JE, Langham ME, Maumenee AE. Manometric studies of the pressure-volume relationship in living and enucleated eyes of individual human subjects. *Br J Ophthalmol*. 1962;46:536–548.
18. Sayah DN, Mazzaferri J, Ghesquiere P, et al. Non-invasive in vivo measurement of ocular rigidity: clinical validation, repeatability and method improvement. *Exp Eye Res*. 2020;190:107831.
19. Sayah DN, Mazzaferri J, Descovich D, et al. The association between ocular rigidity and neuroretinal damage in glaucoma. *Invest Ophthalmol Vis Sci*. 2020;61:11.
20. Sayah DN, Szigiato AA, Mazzaferri J, et al. Correlation of ocular rigidity with intraocular pressure spike after intravitreal injection of bevacizumab in exudative retinal disease. *Br J Ophthalmol*. 2021;105:392–396.
21. Sayah DN, Mazzaferri J, Descovich D, et al. Ocular rigidity and neuroretinal damage in patients with vasospasticity: a pilot study. *Can J Ophthalmol*. 2022;58:338–345.
22. Sayah DN, Descovich D, Costantino S, Lesk MR. The association between the pulsatile choroidal volume change and ocular rigidity. *Ophthalmol Sci*. 2024;4:100576.
23. Masis Solano M, Belanger Nzakimuena C, Dumas R, et al. Ocular rigidity and choroidal thickness changes in response to microgravity: a case study. *Am J Ophthalmol Case Rep*. 2023;32:101940.
24. Bullimore MA, Brennan NA. Myopia control: why each diopter matters. *Optom Vis Sci*. 2019;96:463–465.
25. Sayah DN, Lesk MR. Ocular rigidity and current therapy. *Curr Eye Res*. 2022;48:105–113.
26. Dastiridou AI, Ginis H, Tsilimbaris M, et al. Ocular rigidity, ocular pulse amplitude, and pulsatile ocular blood flow: the effect of axial length. *Invest Ophthalmol Vis Sci*. 2013;54: 2087–2092.
27. Wang J, Freeman EE, Descovich D, et al. Estimation of ocular rigidity in glaucoma using ocular pulse amplitude and pulsatile choroidal blood flow. *Invest Ophthalmol Vis Sci*. 2013;54: 1706–1711.
28. Sayah DN. *Ocular rigidity: a previously unexplored risk factor in the pathophysiology of open-angle glaucoma: assessment using a novel OCT-based measurement method. papyrus.bib.umontreal.ca*. Montreal, Canada: Université de Montréal; 2020.
29. Jonas JB, Ang M, Cho P, et al. IMI prevention of myopia and its progression. *Invest Ophthalmol Vis Sci*. 2021;62:6.
30. Pusti D, Patel NB, Ostrin LA, et al. Peripheral choroidal response to localized defocus blur: influence of native peripheral aberrations. *Invest Ophthalmol Vis Sci*. 2024;65:14.
31. Mangouritsas G, Mourtzoukos S, Mantzounis A, Alexopoulos L. Comparison of Goldmann and Pascal tonometry in relation to corneal hysteresis and central corneal thickness in non-glaucomatous eyes. *Clin Ophthalmol*. 2011;5:1071–1077.
32. Andreanos K, Koutsandrea C, Papaconstantinou D, et al. Comparison of Goldmann applanation tonometry and Pascal dynamic contour tonometry in relation to central corneal thickness and corneal curvature. *Clin Ophthalmol*. 2016;10: 2477–2484.
33. Ozcura F, Yildirim N, Tambova E, Sahin A. Evaluation of Goldmann applanation tonometry, rebound tonometry and dynamic contour tonometry in keratoconus. *J Optom*. 2017;10: 117–122.
34. Bland JM, Altman DG. Measurement error. *BMJ*. 1996;313: 744.
35. Guerrier G, Abdoul H, Jilet L, et al. Efficacy of a web app-based music intervention during cataract surgery: a randomized clinical trial. *JAMA Ophthalmol*. 2021;139:1007–1013.
36. Bray L, Booth L, Gray V, et al. Interventions and methods to prepare, educate or familiarise children and young people for radiological procedures: a scoping review. *Insights Imaging*. 2022;13:146.
37. Kuo AN, Verkicharla PK, McNabb RP, et al. Posterior eye shape measurement with retinal OCT compared to MRI. *Invest Ophthalmol Vis Sci*. 2016;57:OCT196-203.